



CENG 302

Spring 2024-2025

Homework 3 - SQL

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Task Solutions

During creating tables and inserting values into the tables, the main problem is that in Character table, there are two guildId values of 0. However, there is no guildId value of 0. Thus, owing to the guildId values in the fourth and fifth rows of Character table, inserting values into tables was not completed, and SQL statements (SELECT FROM's) cannot be run. After adding the fifth question's answer, the problem was solved. The error is below:

```
ERROR: insert or update on table "character" violates foreign key constraint
"character_guildid_fkey"
Key (guildid)=(0) is not present in table "guild".
```

SQL state: 23503

Detail: Key (guildid)=(0) is not present in table "guild".

- ```
SELECT P.playerId
FROM Player P, Character C, Guild G
WHERE P.playerId=C.playerId AND C.guildId=G.guildId AND G.guildId=1
INTERSECT
SELECT P.playerId
FROM Player P, Character C, Guild G
WHERE P.playerId=C.playerId AND C.guildId=G.guildId AND G.guildId=2;
```

There is no player that have characters in every guild. Cevdet's (11) characters (111 and 116) are in guild 2. Seyit's (12) character (112) is in guild 1. Arthas' (13) characters (114 and 115) are not in a guild. Hekimoğlu's (14) character (113) is in guild 2.

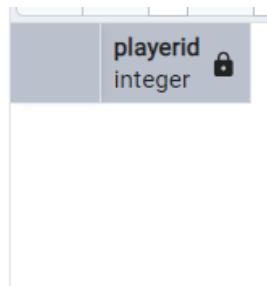


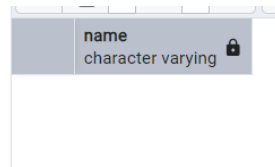
Figure 1: Output of the 1<sup>st</sup> Question

```

2. SELECT P3.name
 FROM Player P3
 WHERE P3.playerId IN ((SELECT C.playerId
 FROM Character C
 WHERE C.class='Mage')
 INTERSECT
 (SELECT C2.playerId
 FROM Character C2
 WHERE C2.class='Rogue')));

```

There is no player that have at least one character of class "Mage" and at least one character of class "Rogue". Arthas' (13) character (115) has class "Mage". Cevdet's (11) character (111) and Seyit's (12) character (112) have class "Rogue". Hence, there is no intersection.



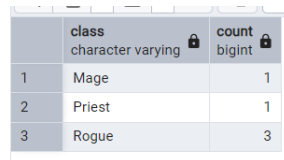
| name | character varying |
|------|-------------------|
|------|-------------------|

Figure 2: Output of the 2<sup>nd</sup> Question

```

3. SELECT C.class, COUNT(QP.status)
 FROM Character C, QuestParticipation QP
 WHERE C.charId=QP.charId AND QP.status='Completed'
 GROUP BY C.class;

```



|   | class             | count  |
|---|-------------------|--------|
|   | character varying | bigint |
| 1 | Mage              | 1      |
| 2 | Priest            | 1      |
| 3 | Rogue             | 3      |

Figure 3: Output of the 3<sup>rd</sup> Question

```

4. SELECT Q.questId, COUNT(DISTINCT C.charId)
 FROM Quest Q, Character C, QuestParticipation QP
 WHERE C.charId=QP.charId AND QP.questId=Q.questId AND Q.questId IN(
 SELECT Q.questId
 FROM Quest Q
 WHERE Q.difficulty='3/5')
 GROUP BY Q.questId
 HAVING COUNT(DISTINCT C.charId) >= 2;

```

|   | questid<br>[PK] integer | count<br>bigint |
|---|-------------------------|-----------------|
| 1 | 70202                   | 2               |

Figure 4: Output of the 4<sup>th</sup> Question

5. `INSERT INTO Guild(guildId, guildName, creationDate) VALUES (0, NULL, NULL);`

*The entire code block is in the URL of  
[https://github.com/mertyucel02/ceng302\\_hw3\\_2515401](https://github.com/mertyucel02/ceng302_hw3_2515401)*